• COMPUTER NETWORKS

Sending Messages: Packet Switching

Learning Objective

After this lesson, you will understand:

- What packet switching is
- How messages travel across the Internet
- Why messages are split into packets
- What information each packet contains
- How the receiver rebuilds the original message

INTRODUCTION

When we send an email, WhatsApp message, or open a website, the message is **not sent as one large file**.

Instead, the Internet breaks the message into many **small pieces** called **packets**.

Each packet travels through the network and reaches the destination.

Finally, the receiver puts all packets back together to recreate the original message.

This process is called Packet Switching.

What is a Packet?

A **packet** is a small piece of a complete message.

ORIGINAL MESSAGE

"Hello Omar, How are you?"

Packet 1 → Hello

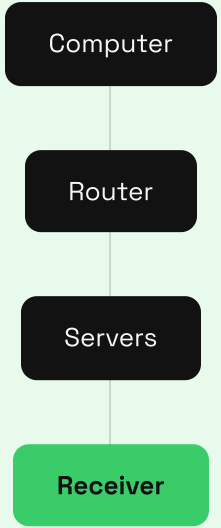
Packet 2 → Omar,

Packet 3 → How

Packet 4 → are you?

How Packets Travel

Different packets can take different paths across the Internet.



Why Different Paths?

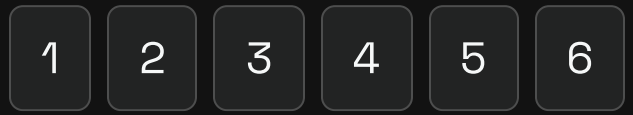
The Internet chooses the best available route.

Faster	Reliable	Efficient

Do Packets Arrive Together?

No.

SENT



RECEIVED

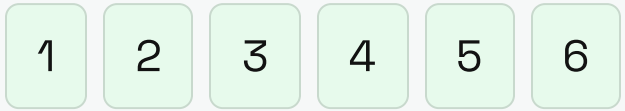


This is normal.

How Does the Receiver Rebuild the Message?

Every packet has a **Sequence Number**.

The receiver sorts the packets by sequence number to rebuild the original message.



What Does a Packet Contain?

1

Destination Address

Where the packet should go.

2

Sequence Number

The correct order of the packet.

3

Total Number of Packets

Shows how many packets belong to the message.

What Happens if a Packet is Lost?



If one packet is missing, the receiver requests **only the missing packet**.

The whole message does not need to be sent again.

Advantages of Packet Switching

- Fast
- Reliable
- Efficient
- Uses different routes
- Only missing packets are resent

Real-Life Examples

- Email
- WhatsApp
- Video Streaming
- Online Gaming
- Web Browsing
- File Downloads

Key Terms

Term	Meaning
Packet	A small piece of data
Router	Connects devices to the Internet
Server	Forwards or provides data
Packet Switching	Sending data in small packets
Sequence Number	Keeps packets in order
Destination Address	Shows where the packet goes

Summary

- Messages are divided into packets.
- Packets travel through routers and servers.
- Different packets can take different routes.
- Packets may arrive out of order.
- Sequence numbers rebuild the message.
- Missing packets are resent.
- This process is called Packet Switching.

Interview Questions

01 What is packet switching?

02 Why are messages divided into packets?

03 Can packets take different routes?

04 Why do packets have sequence numbers?

05 What happens if a packet is lost?

• 30-SECOND EXPLANATION

When we send data over the Internet, the computer divides it into small packets. Each packet can travel through different routes. At the destination, the receiver uses sequence numbers to put the packets back into the correct order. If one packet is lost, only that packet is sent again. This process is called Packet Switching.